ETALE MORPHISMS ONTO A MCQUILLAN AFFINE FORMAL ALGEBRAIC SPACE ARE FORMAL SPECTRA OF COMPLETIONS OF ETALE ALGEBRAS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one chain of identifications of colimits; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The formal spectrum functor on complete weakly admissible algebras is fully faithful. Same chapter, Lemma lemma-morphism-between-formal-spectra, invoked at proof lines 4464-4466. Gives full faithfulness of the functor constructed in the statement.
- A representable etale morphism of affine formal algebraic spaces has affine scheme fibres. Same chapter, Lemma lemma-affine-representable-by-algebraic-spaces, invoked at proof lines 4472-4474. Makes the base change of the morphism to a quotient an etale morphism of affine schemes.
- An etale algebra map over a quotient can be lifted along the quotient map. Algebra, Lemma algebra-lemma-lift-etale, invoked at proof lines 4475-4477. Produces the etale B-algebra whose completion is the candidate source.
- Uniqueness of etale lifts along a locally nilpotent ideal. More on Algebra, Lemma more-algebra-lemma-locally-nilpotent-henselian, invoked at proof lines 4481-4484. Gives the compatible isomorphisms between the two lifts, which is what identifies the given space with a formal spectrum.

Source reference. Original lemma 04D1: sources/stacks-chow/tags.txt, source lines 5711–5711. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0AN0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2355–2355. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ALI: sources/stacks-chow/tags.txt, source lines 13371–13371. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let Y be a McQuillan affine formal algebraic space over S, i.e., $Y = Spf(B)$ for some weakly admissible topological S-algebra B. Then there is an equivalence of categories between*

1. *the category of morphisms $f : X \to Y$ of affine formal algebraic spaces which are representable by algebraic spaces and étale, and*
2. *the category of topological B-algebras of the form $A^\wedge$ where A is an étale B-algebra and $A^\wedge = \lim A/JA$ with $J \subset B$ running over the weak ideals of definition of B.*

The equivalence is given by sending $A^\wedge$ to $X = Spf(A^\wedge)$. In particular, any X as in (1) is McQuillan.

Proof. Let A be an étale B-algebra. Then $B/J \to A/JA$ is étale for every open ideal $J \subset B$. Hence the morphism $\mathrm{Spf}(A^\wedge) \to Y$ is representable and étale. The functor Spf is fully faithful by Lemma 0AN0. To finish the proof we will show in the next paragraph that any $X \to Y$ as in (1) is in the essential image.

Choose a weak ideal of definition $J_0 \subset B$. Set $Y_0 = \mathrm{Spec}(B/J_0)$ and $X_0 = Y_0 \times_Y X$. Then $X_0 \to Y_0$ is an étale morphism of affine schemes (see Lemma 0AKN). Say $X_0 = \mathrm{Spec}(A_0)$. By Algebra, Lemma 04D1 we can find an étale algebra map $B \to A$ such that $A_0 \cong A/J_0A$. Consider an ideal of definition $J \subset J_0$. As above we may write $\mathrm{Spec}(B/J) \times_Y X = \mathrm{Spec}(\bar{A})$ for some étale ring map $B/J \to \bar{A}$. Then both $B/J \to \bar{A}$ and $B/J \to A/JA$ are étale ring maps lifting the étale ring map $B/J_0 \to A_0$. By More on Algebra, Lemma 0ALI there is a unique B/J-algebra isomorphism $\varphi_J : A/JA \to \bar{A}$ lifting the identification modulo J_0. Since the maps φ_J are unique they are compatible for varying J. Thus

$$X = \mathrm{colim}\, \mathrm{Spec}(B/J) \times_Y X = \mathrm{colim}\, \mathrm{Spec}(A/JA) = \mathrm{Spf}(A)$$

and we see that the lemma holds. □